

Aidan Ream

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Overview

Computer Graphics Programmer with experience in both high-level applications and low-level implementation. Experienced with real time rendering, performance optimizations, and visual design choices.

Projects

Monte Carlo Ray Tracer

- Made a C++ Path Tracer with the following features: Geometric primitive and triangle-mesh support, performance optimization structures, multi-threading, png output, Importance Sampling, Atmospheric Persepective camera effects, and more.
- Tools Used: C++

Compute Shader Based Vulkan Cloth Simluation

- Created a real time simulation environment with the Vulkan graphics library that utilizes a spring based compute shader on a grid of connected points to simulate the movement and flow of cloth using the GPU for performance optimization.
- Tools Used: C++, Vulkan

Skeletal Animation System For Custom Game Engine

- Collaborated on a 3D Roguelike adventure game created by building a custom game engine with OpenGL. My primary contributions were implementing a skeletal animation system, 3D modeling and animations, and creating the debug mode.
- Tools Used: C++, openGL, Blender

Dynamic Dithering Shader

- Created a Shader Algorithm to introduce “dithering” printing effect to a dynamic 3D Scene
- Tools Used: C++, openGL, GLSL

Technology Demo for Expressive Technology Studios

- Built an immersive technology demo in Unity combining motion capture, blender animation, and spatial audio.
- Tools Used: Unity, Blender, Rokoko studio, Davinci Resolve

Experience

Calpoly, Lab Manager at Expressive Technology Studios

- Provided software updates and equipment upkeep for lab equipment over the course of 2.5 years
- Ran 10+ events per year utilizing our spatial audio theater space, assisting visiting speakers with the use of the software

San Luis Obispo, CA
August 2023 – December 2025

ACM SIGGRAPH, Student Volunteer

- Assisted conference attendees with navigation and scheduling
- Provided presenter assistance during four performances of 'Spatial Storytelling'

Vancouver, BC - August 2025
Los Angeles, CA - July 2026

Skills

Programming Languages: C++, C, Java, Python, GLSL

Graphics Libraries & Game Engines: openGL, Vulkan, Unity, Godot, Houdini, RenderDoc

Visual Design Software: Maya, Blender, Substance Painter, Photoshop, Adobe Illustrator, Krita

Other Skills: Agile/Scrum workflows, Git source control, Group communication, Team management

Education

Calpoly, San Luis Obispo CA
Bachelor of Science in Liberal Arts and Engineering Studies

Graduated June 2026
Cumulative GPA 3.54

Computer Graphics concentration - Coursework focused on computer graphics projects

2D and 3D Animation concentration - Coursework focused on design principles and industry standard workflows/technologies

Computer Science Minor - Additional coursework taken exploring various computer science disciplines